

- 1) Use the first principle of derivative to find the derivative of $g(x) = 3x^2 + 4x - 2$. Show step of calculation.

Given,

$$g(x) = 3x^2 + 4x - 2$$

$$g(x+h) = 3(x+h)^2 + 4(x+h) - 2$$

Using first principle of derivative,

$$g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3(x+h)^2 + 4(x+h) - 2 - (3x^2 + 4x - 2)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3(x^2 + 2hx + h^2) + 4x + 4h - 2 - 3x^2 - 4x + 2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{3x^2} + 6hx + 3h^2 + \cancel{4x} + 4h - \cancel{2} - \cancel{3x^2} - \cancel{4x} + \cancel{2}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6hx + 3h^2 + 4h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(6x + 3h + 4)}{h}$$

$$= 6x + 3 \times 0 + 4$$

$$= 6x + 4$$

$$\therefore g'(x) = 6x + 4$$

2) The position function of an object moving along a straight line is given by $S(t) = 2t^2 - 3t + 1$. Find the velocity of an object at time $t = 2$ using the concept of limits and tangent lines.

Given,

$$S(t) = 2t^2 - 3t + 1$$

$$S(t+h) = 2(t+h)^2 - 3(t+h) + 1$$

Using first principle of derivatives,

$$S'(t) = \lim_{h \rightarrow 0} \frac{S(t+h) - S(t)}{h}$$

$$S'(t) = \lim_{h \rightarrow 0} \frac{2(t+h)^2 - 3(t+h) + 1 - (2t^2 - 3t + 1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2(t^2 + 2ht + h^2) - 3t - 3h + 1 - 2t^2 + 3t - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2t^2 + 4ht + 2h^2 - 3t - 3h + 1 - 2t^2 + 3t - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{4ht + 2h^2 - 3h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(4t + 2h - 3)}{h}$$

$$= \lim_{h \rightarrow 0} 4t + 2h - 3$$

$$= 4t + 2 \times 0 - 3$$

$$= 4t - 3$$

$$\therefore S'(t) = 4t - 3$$

To find velocity of an object at time $t=2$

$$S'(2) = 4 \times 2 - 3 = 8 - 3 = 5$$

$\therefore$ The velocity of an object at time is 5 units per second

3) The population of a city is modeled by $P(t) = 1200e^{0.02t}$ where t is the time in years. Find the rate at which the population is growing after 3 years. Interpret the result in the context of the problem.

Given,

$$P(t) = 1200e^{0.02t}$$

$$P(t+h) = 1200e^{0.02(t+h)}$$

Using first principle of derivative,

$$P'(t) = \lim_{h \rightarrow 0} \frac{P(t+h) - P(t)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1200e^{0.02t+0.02h} - 1200e^{0.02t}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1200 e^{0.02t} (e^{0.02h} - 1)}{h}$$

Multiplying numerator and denominator by 0.02

$$\lim_{h \rightarrow 0} \frac{0.02 \times 1200 e^{0.02t} (e^{0.02h} - 1)}{0.02 h}$$

$$\lim_{h \rightarrow 0} \frac{24 e^{0.02t} (e^{0.02h} - 1)}{0.02h}$$

We know, $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$ also in the question it is in the form $\lim_{h \rightarrow 0} \frac{e^{0.02h} - 1}{0.02h}$

We can write it as 1

$$= 24 e^{0.02t} \times 1$$

$$= 24 e^{0.02t}$$

$$\therefore P'(t) = 24 e^{0.02t}$$

To find the rate of population growing after 3 years put $t = 3$

$$\therefore P'(3) = 24 e^{0.02 \times 3} = 24 e^{0.06} = 25.48$$

This means that after 3 years, the city's population is increasing at a rate of about 25 people per year. So, at the moment in time, the population grows continuously and the number of people added each year is approximately 25.

$$S'(t) = 4 \times 20 = 80 \text{ units per second}$$

The velocity of an object at time t is 3 units per second.

3) The population of a city is modeled by $P(t) = 1200e^{0.02t}$, where t is the time in years. Find the rate at which the population is growing at $t = 5$ years. Interpret the result in the context of the problem.

Given: $P(t) = 1200e^{0.02t}$

$$P(t) = 1200e^{0.02t}$$

$$P(t+h) = 1200e^{0.02(t+h)}$$

Using the principle of differentiation

$$P'(t) = \lim_{h \rightarrow 0} \frac{P(t+h) - P(t)}{h}$$

$$h \rightarrow 0 \quad t \quad h$$

$$\lim_{h \rightarrow 0} \frac{1200e^{0.02(t+h)} - 1200e^{0.02t}}{h}$$

$$1200e^{0.02t} \cdot 0.02$$